Unit 11, Lesson 1: Representing Bivariate Data

Benchmark

MA.8.DP.1.1 Given a set of real-world bivariate numerical data, construct a scatter plot or a line graph as appropriate for the context.

Learning Targets

- ☐ **I can determine the independent and dependent variables in a given set of real-world bivariate data.**
- ☐ **I can determine if the data is best represented by a line graph or scatter plot.**
- ☐ **I can determine an appropriate scale for the axis representing each variable.**

11.1.1 Warm-Up: Army Dentist

When Captain Ryan Romero was asked how he got started as an Army dentist, he said his first job was to be a dental assistant. He was sitting on the other side of the chair helping the dentist and he enjoyed it so much that he decided to pursue a career in Army dentistry.

1. **When thinking about your future career, it's helpful to explore areas of interest to help you identify careers you might like. What is a career you think you would like to have?**

2. **What could you do now or in the future to help you determine if that is a career you would enjoy?**

11.1.2 Guided Instruction: Representing Bivariate Data

1. The U.S. Centers for Disease Control and Prevention (CDC) is the nation's health protection agency. Epidemiologists at the CDC study the spread and impact of diseases. The table shows the number of symptomatic cases of influenza, also called the flu, in millions, in the U.S. since 2011.

Years Since 2011	0	1	2	3	4	5	6	7	8	9
Number of Cases, in millions	21	9.3	34	30	30	24	29	45	36	38

a. What year does the number 6 represent?

b. Keanu and Francisca were discussing what they noticed about the values in the table.

Keanu	Francisca
I noticed the actual year was not used in the table. I think this makes sense because the flu may not have existed in 1700.	I noticed the information given in the table stated the information was from a study. I think the study did not start until 2011.

Discuss Keanu's and Francisca's ideas with your partner.

2. Discuss with your partner what you notice in the data. Summarize your discussion.

This data set is an example of *bivariate data* because two variables are being compared.

Bivariate data is data that measures two characteristics of a population.

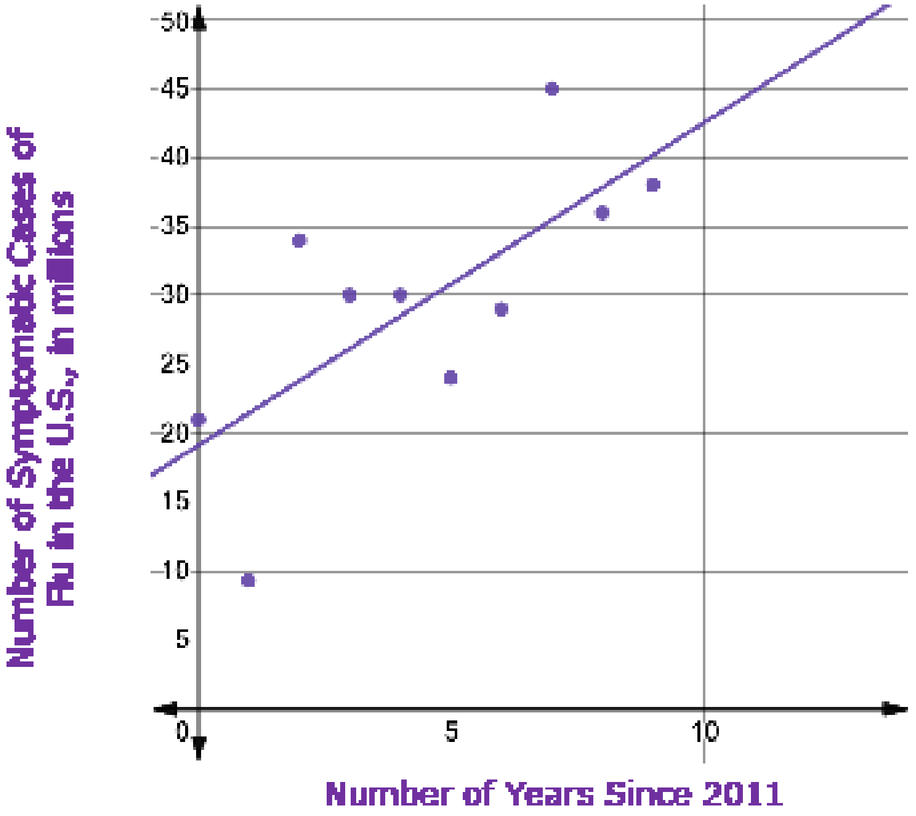
3. Based on the definition of bivariate data, what are the two characteristics measured in the table?

Bivariate numerical data can be represented in different ways. One way to represent the data is a *scatter plot* with a *line of fit*.

A ***scatter plot*** is a graph in the coordinate plane representing a set of bivariate numerical data that is used to observe the relationship between two variables.

A ***line of fit*** is a line drawn on a scatter plot to estimate the relationship between two sets of data. A line of fit is also known as a trend line.

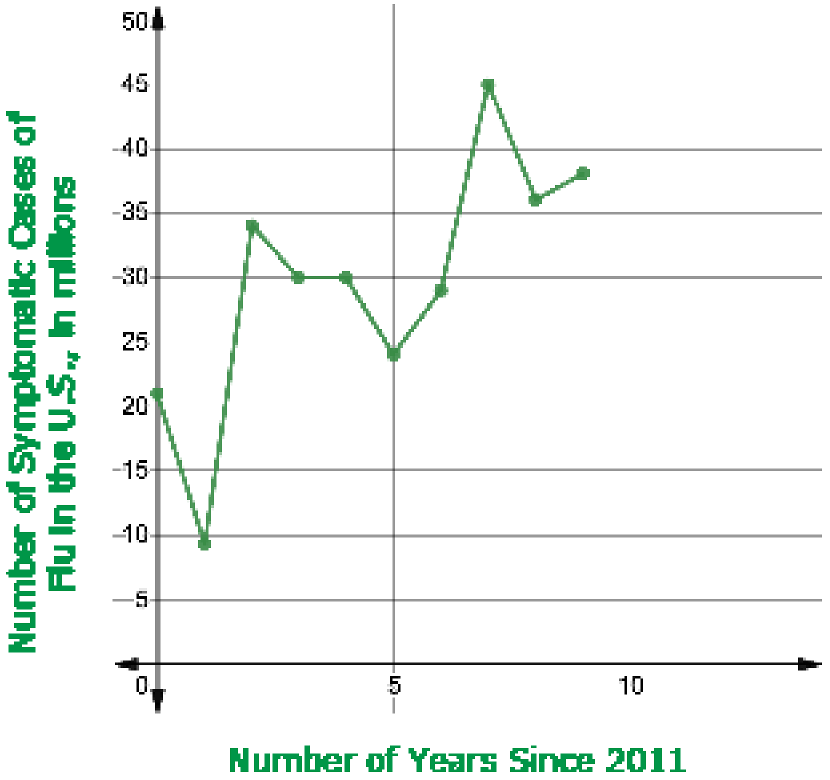
A scatter plot of the data on symptomatic cases of the flu in the U.S. is shown. The line of fit, $y = 2.34x + 19.1$, is included on the scatter plot, where y is the number of symptomatic cases of the flu in the U.S., in millions, and x is the number of years since 2011.



The same data can be represented using a ***line graph***.

A ***line graph*** is a graph that displays numerical data using connected line segments.

A line graph of the data on symptomatic cases of the flu in the U.S. is shown.



4. Discuss with your partner the similarities and differences between the scatter plot and the line graph.

   a. Complete the table.

Similarities Between Scatter Plots and Line Graphs	Differences Between Scatter Plots and Line Graphs

b. **Mateo is interested in determining how the number of flu cases in the U.S. has changed each year since 2011. Explain which graph is best for him to use.**

c. **Londyn is interested in the overall trend in the number of flu cases in the U.S. since 2011. Explain which graph is best for her to use.**

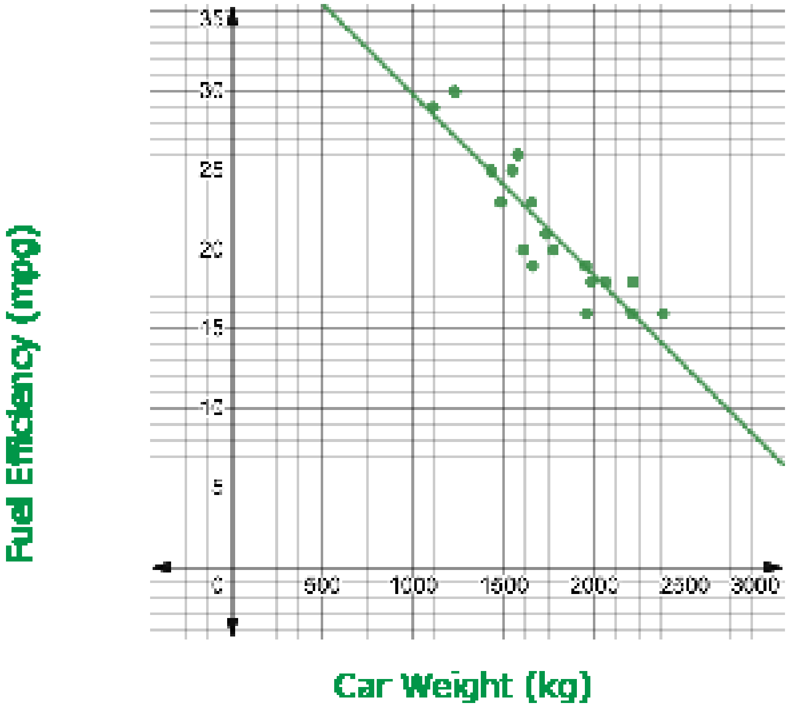
d. **Discuss with your partner which variable is the independent variable and which is the dependent variable in this context. Identify your choices.**

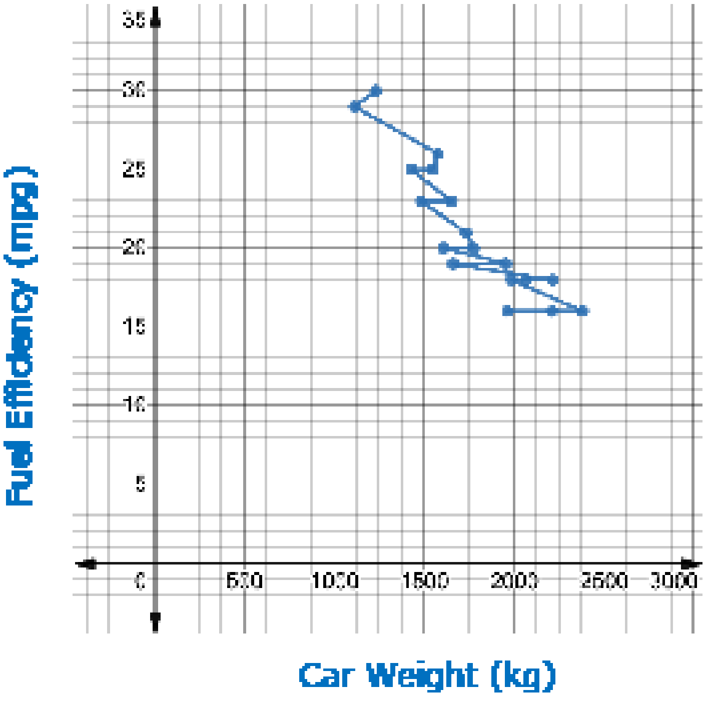
Independent Variable	o Number of cases of the flu o Years since 2011
Dependent Variable	o Number of cases of the flu o Years since 2011

Both scatter plots and line graphs can be used to observe the relationship between two variables in a set of bivariate data. While overall trends can be seen in both graphs, line graphs allow for closer analysis from one point to another in the data set.

11.1.3 Guided Practice: Representing Bivariate Data

1. A scatter plot and a line graph comparing the weights, in kilograms (kg), and fuel efficiencies, in miles per gallon (mpg), of 18 cars are shown.





a. Discuss with your partner which graph you think more clearly shows the relationship between a car's weight and its fuel efficiency. Summarize your discussion.

b. Complete the statement to describe the relationship shown on the graph identified in Part A.

The graph shows that, in general, a car's fuel efficiency

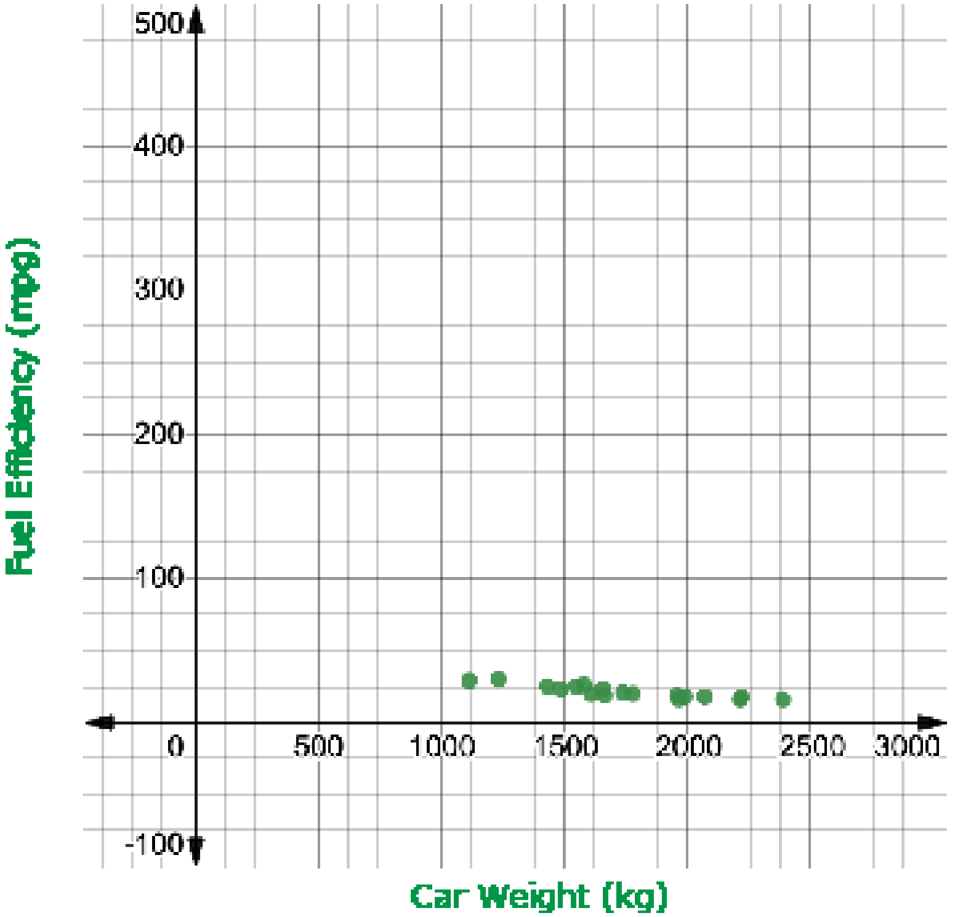
- o increases
- o decreases

as the car's weight increases.

c. Kate noticed that there were cars of two different weights with the same fuel efficiency of 25 mpg. How does this impact the visual representation of the data on the line graph?

When the goal of analyzing a set of bivariate numerical data is to observe the relationship between the two variables overall, a scatter plot is most useful.

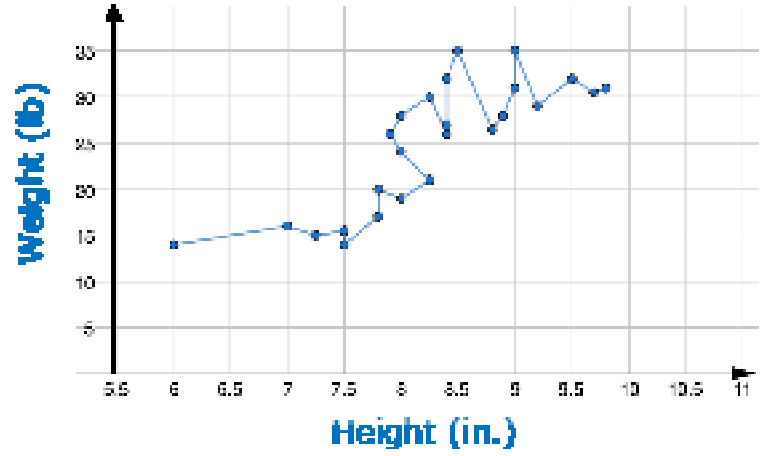
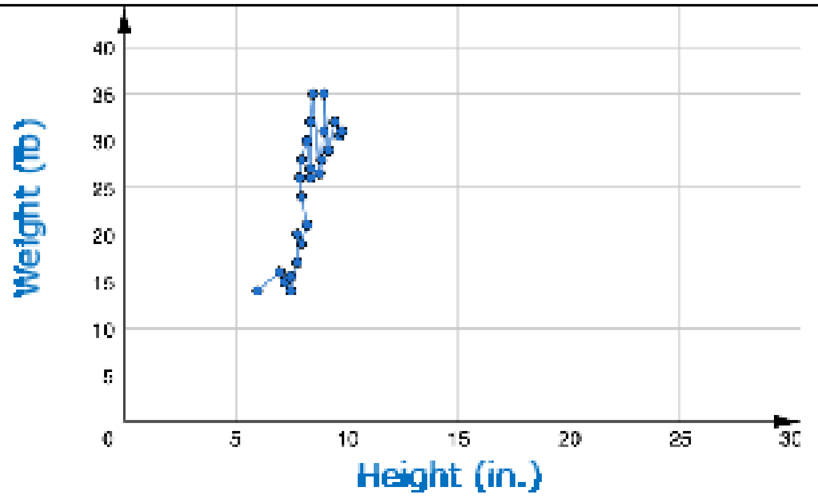
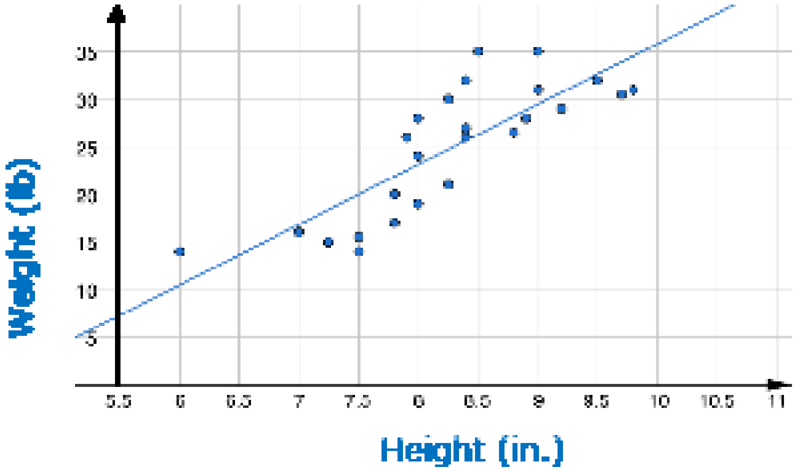
d. **A scatter plot representing the same bivariate data set, but using a different scale is shown.**

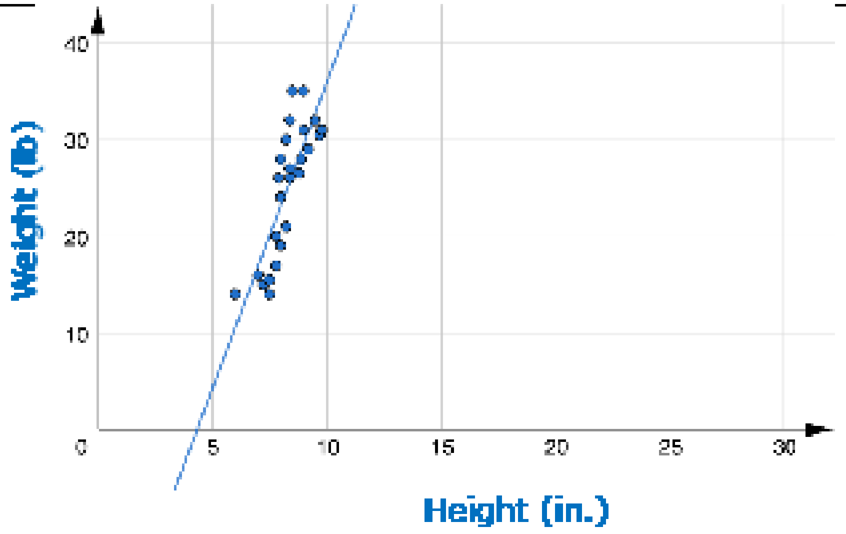


Explain how the change in scale could influence how the relationship between the two variables is interpreted.

11.1.4 Collaboration: Choosing an Appropriate Graph

1. The relationship between the height, in inches (in.), and weight, in pounds (lb), of 25 dachshunds, a breed of dogs, is represented on the graphs.

Line Graph A	
Line Graph B	
Scatter Plot A	

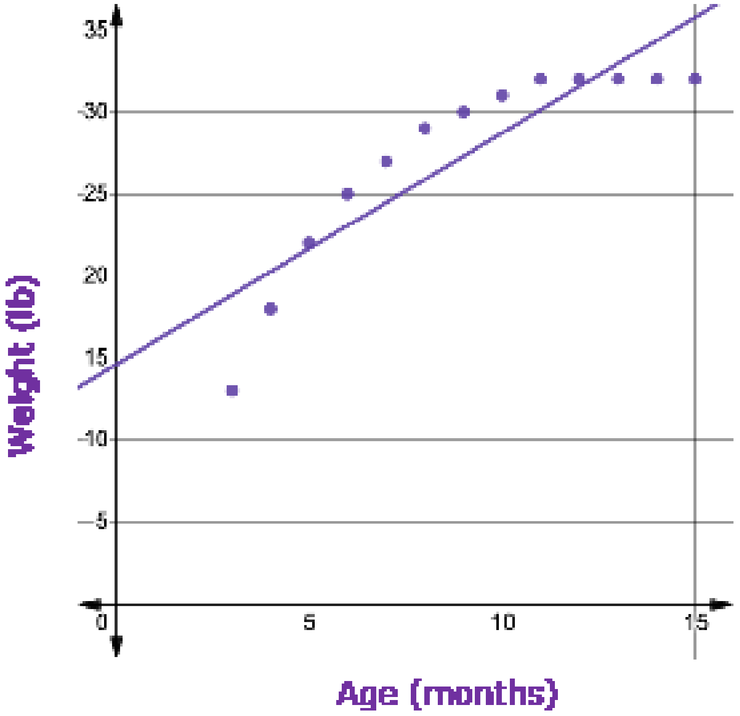
Scatter Plot B	

a. Work with your partner to explain which of the graphs is best to show the relationship between the weight and height of these dogs. Be sure to include information about the type of graph and the scale in your explanation.

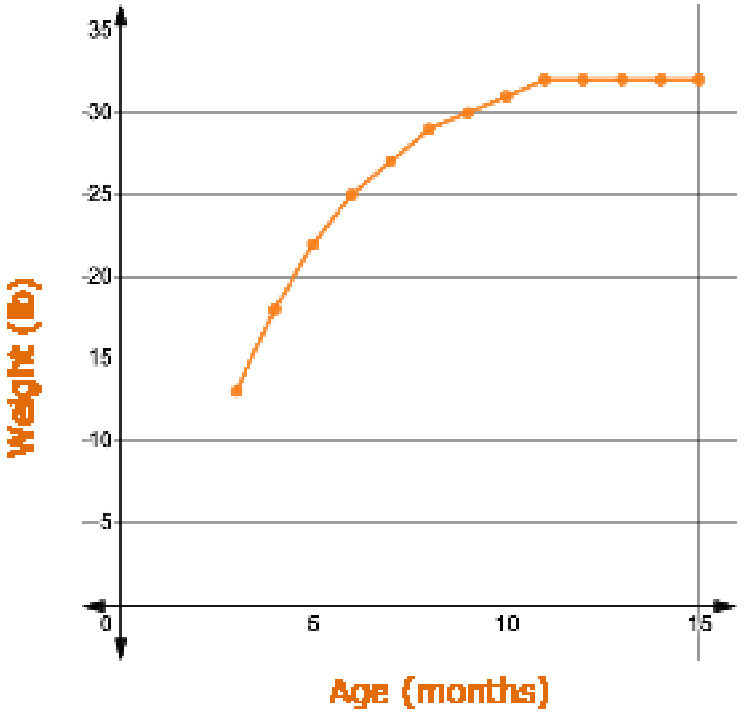
b. Describe the relationship between a dachshund's height and its weight using the graph identified in Part A.

2. The relationship between the age of a dachshund, in months, and its weight, in pounds (lb), is shown on the graphs.

Scatter Plot



Line Graph



Explain which of the graphs best demonstrates the relationship between the age and weight of dachshunds.

11.1.5 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each learning target.

I need help. I got it. I could help someone.

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I can determine if the data is best represented by a line graph or scatter plot.	
I can determine the best scale for the axis for which each variable will be represented.	

